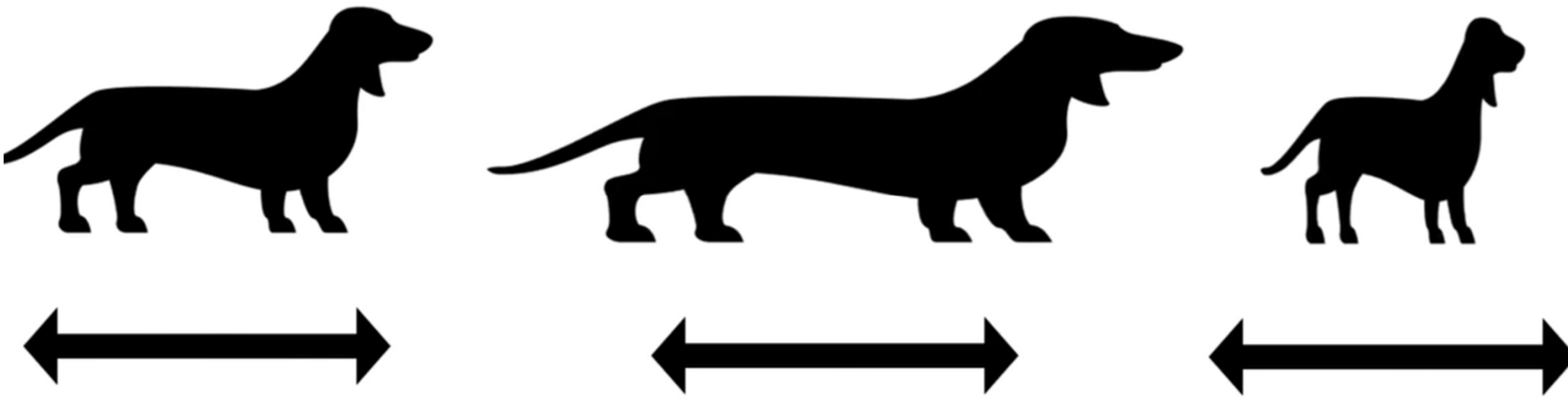
Singular Value Decomposition

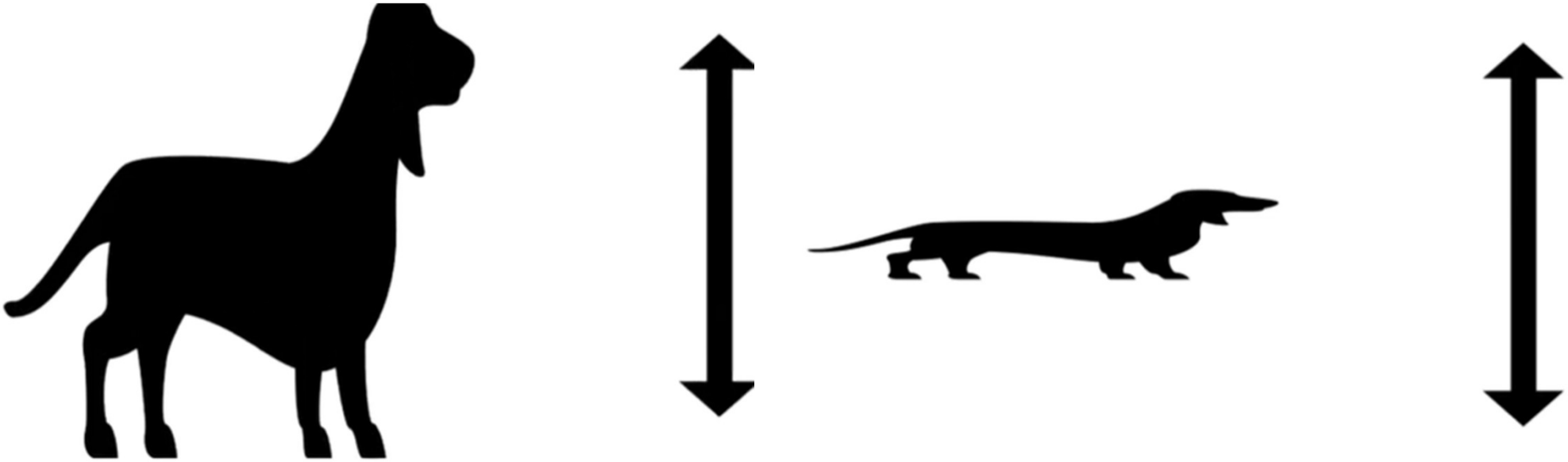
Transformation of matrix

- Stretch or compress horizontally



Transformation

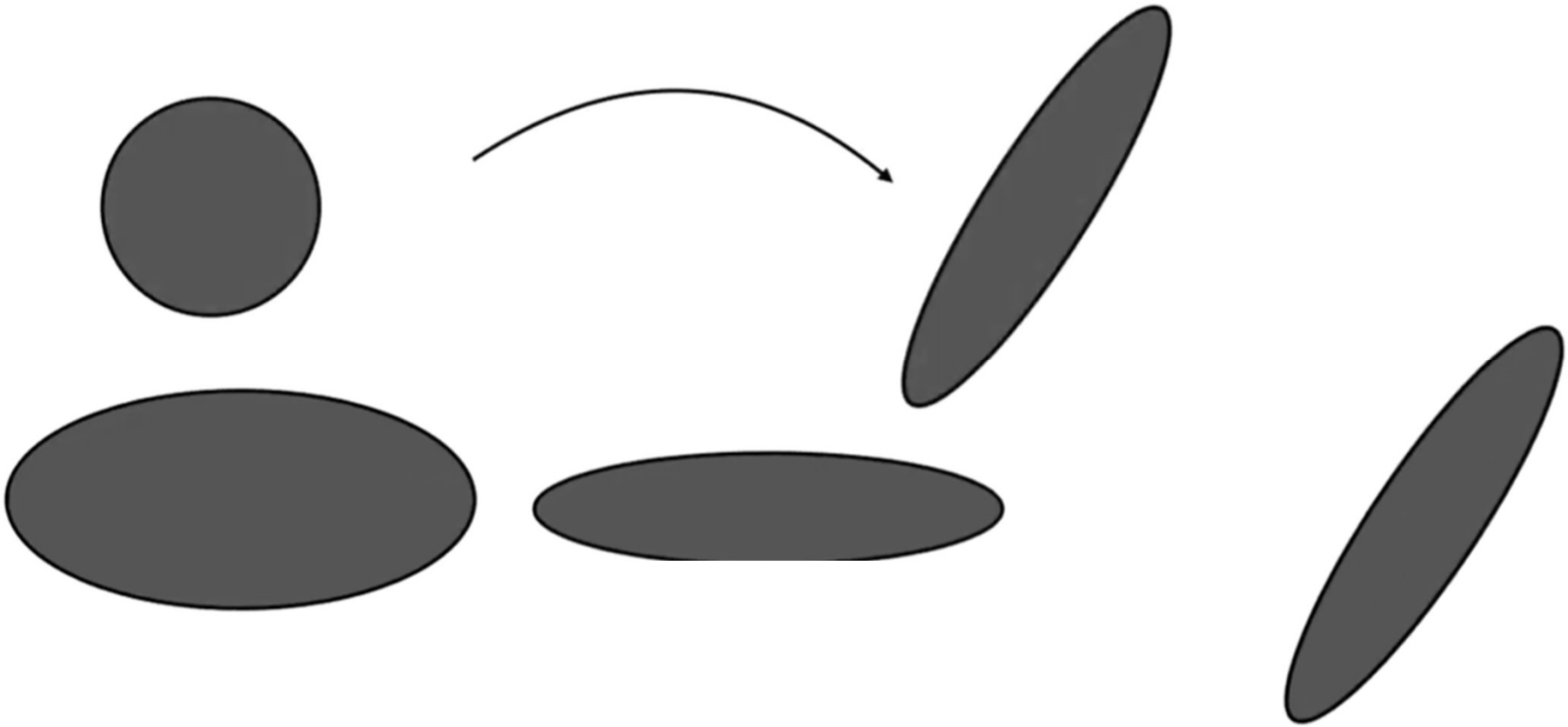
Stretch or compress vertically

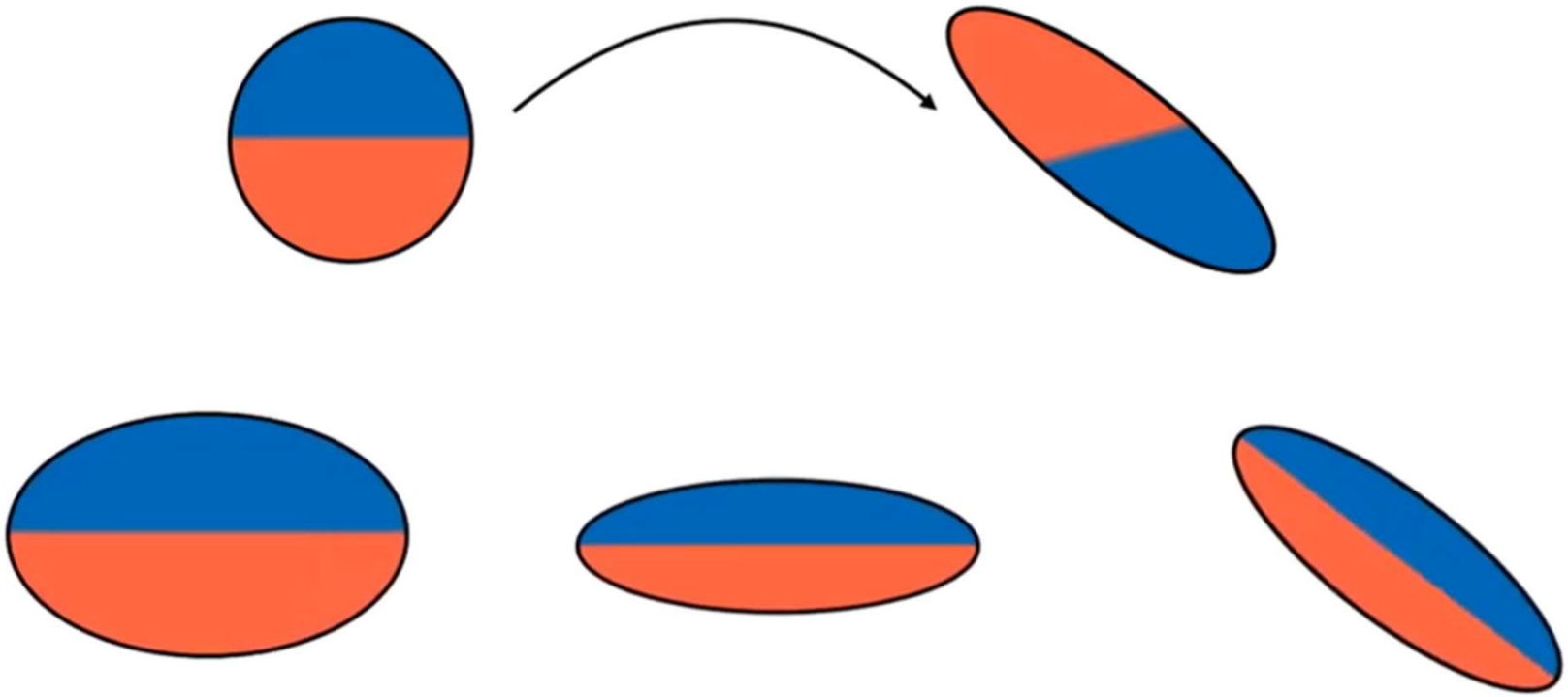


Transformation

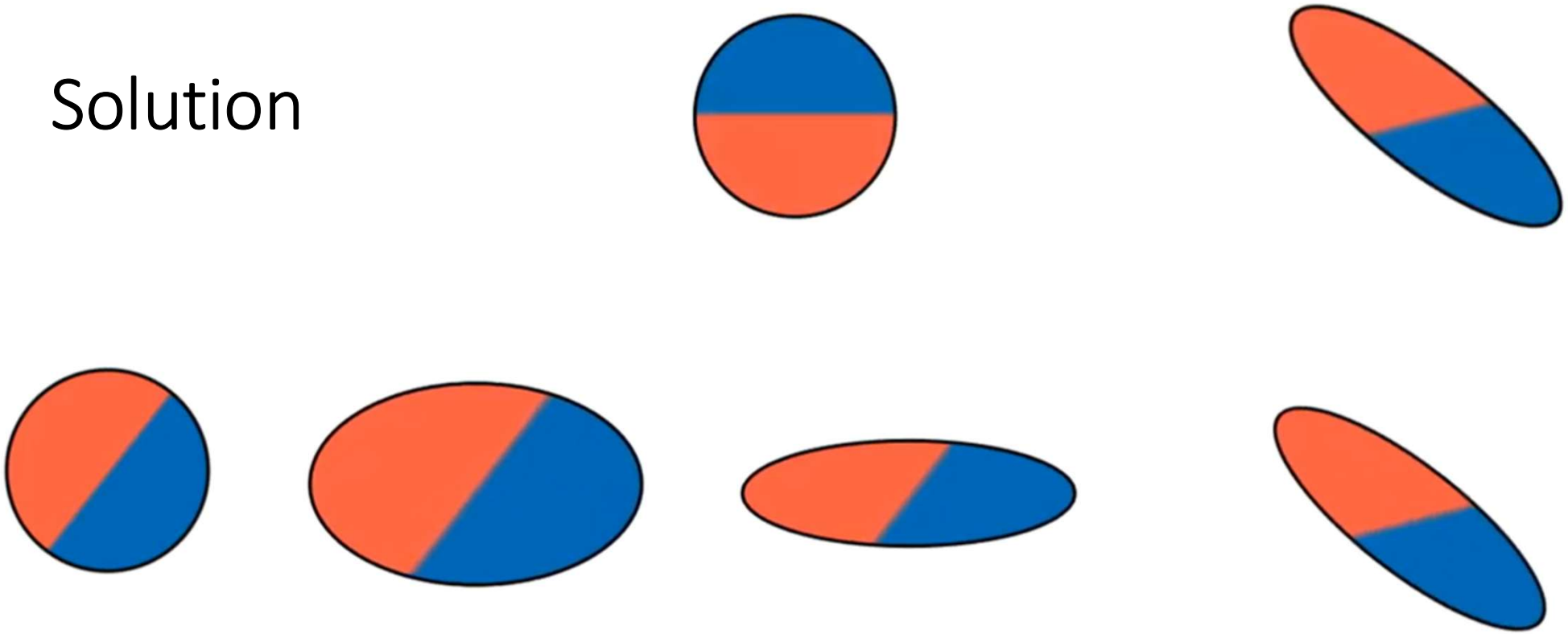
- Rotate

Puzzle(Stretching horizontally and vertically



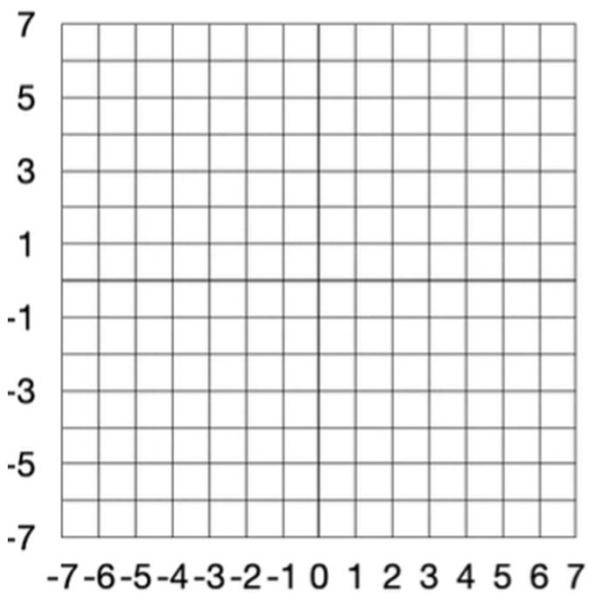


Solution

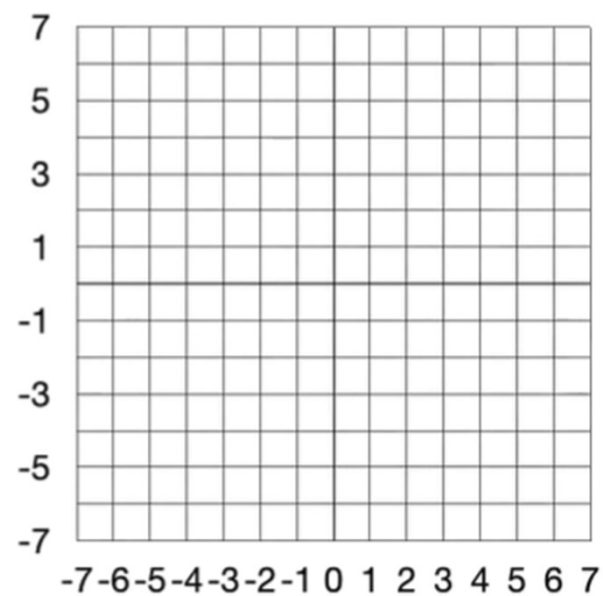


Linear Transformation

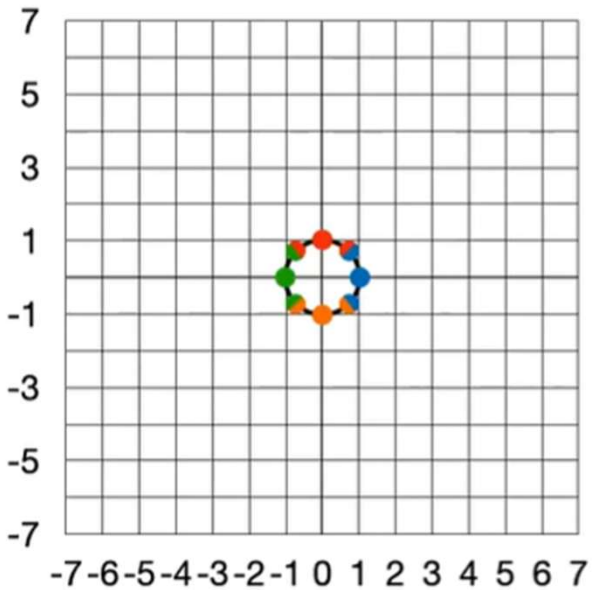
$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$



(p,q)	(3p+0q, 4p+5q)
(1,0)	(3, 4)
(0,1)	(0, 5)
(0,1)	(0, 5)
(-1,0)	(-3, -4)
(0,-1)	(0, -5)

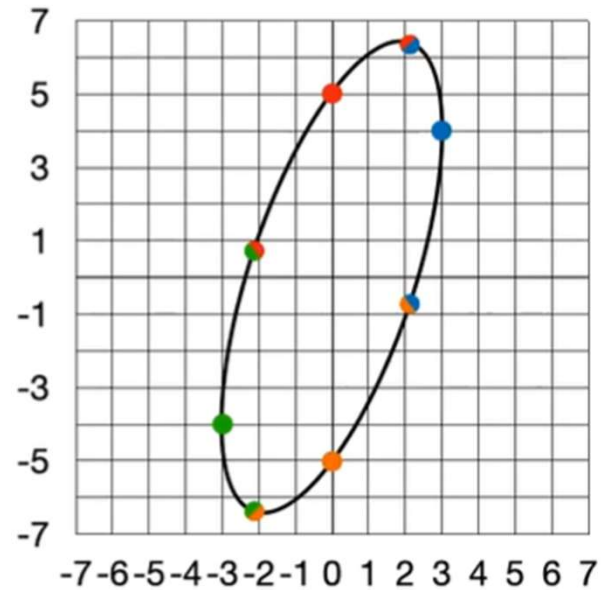


Linear Transformation

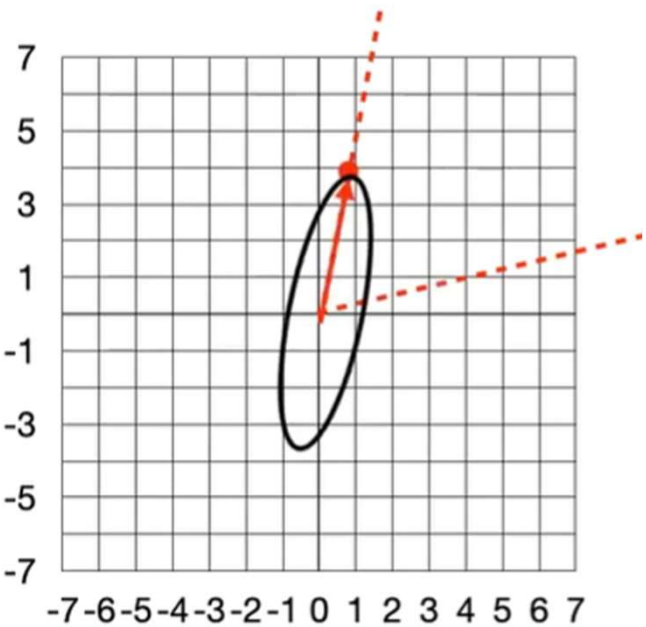


$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$

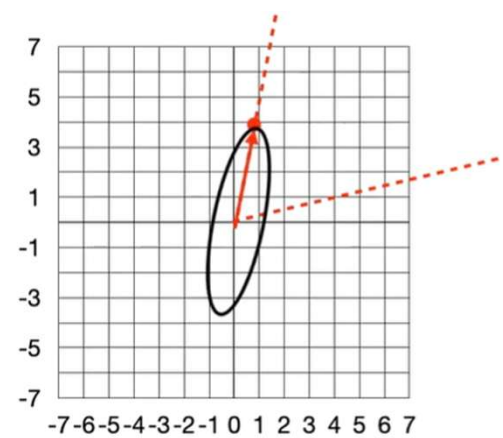
(p,q)	(3p+0q, 4p+5q)
(1,0)	(3, 4)
(0,1)	(0, 5)
(-1,0)	(-3, -4)
(0,-1)	(0, -5)

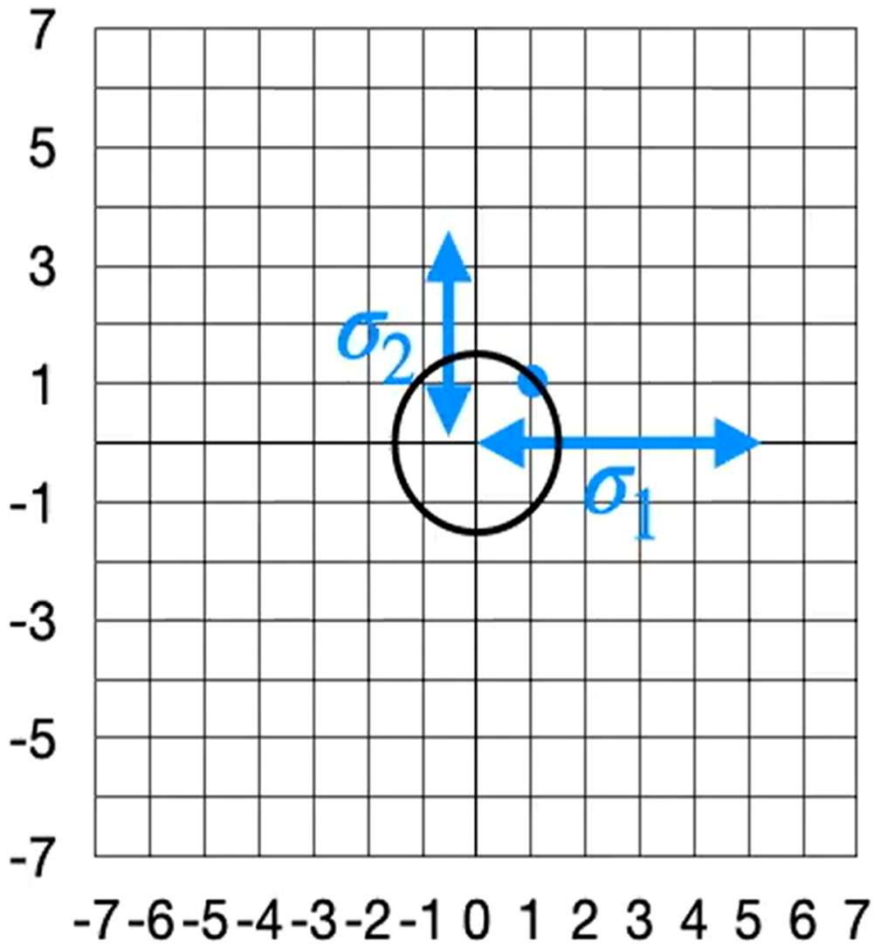


Rotation Matrices



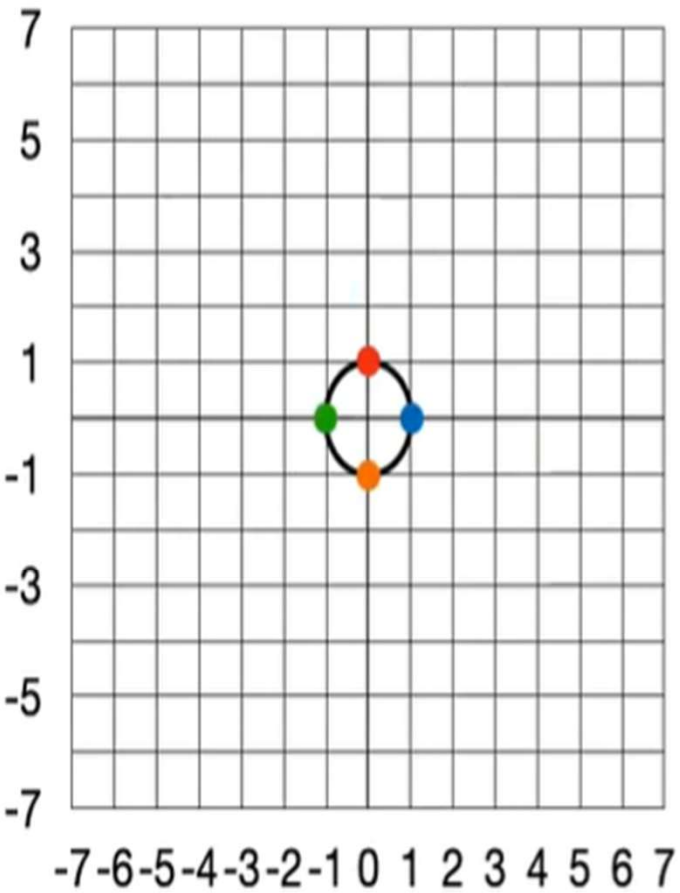
$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

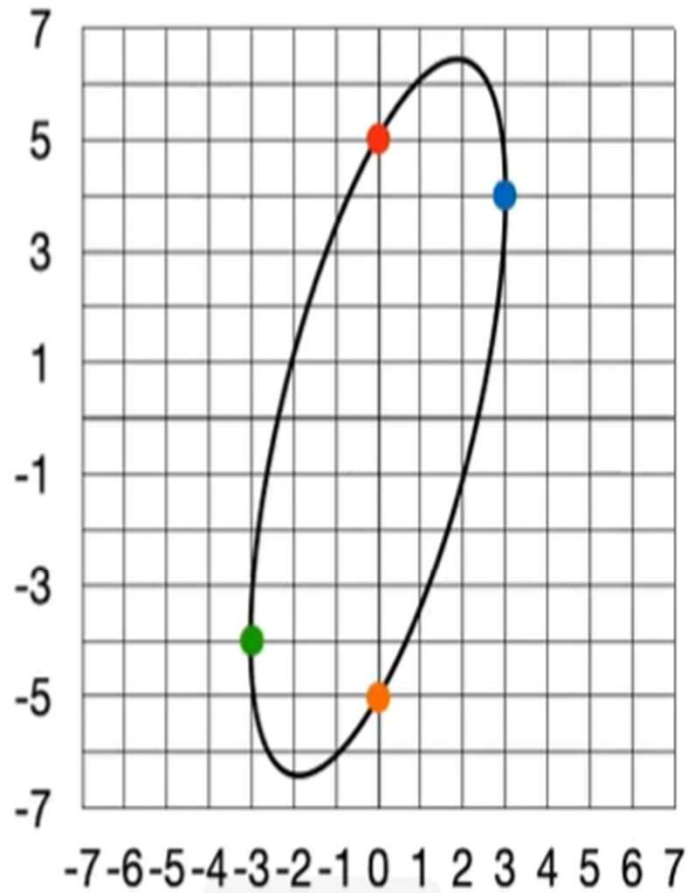




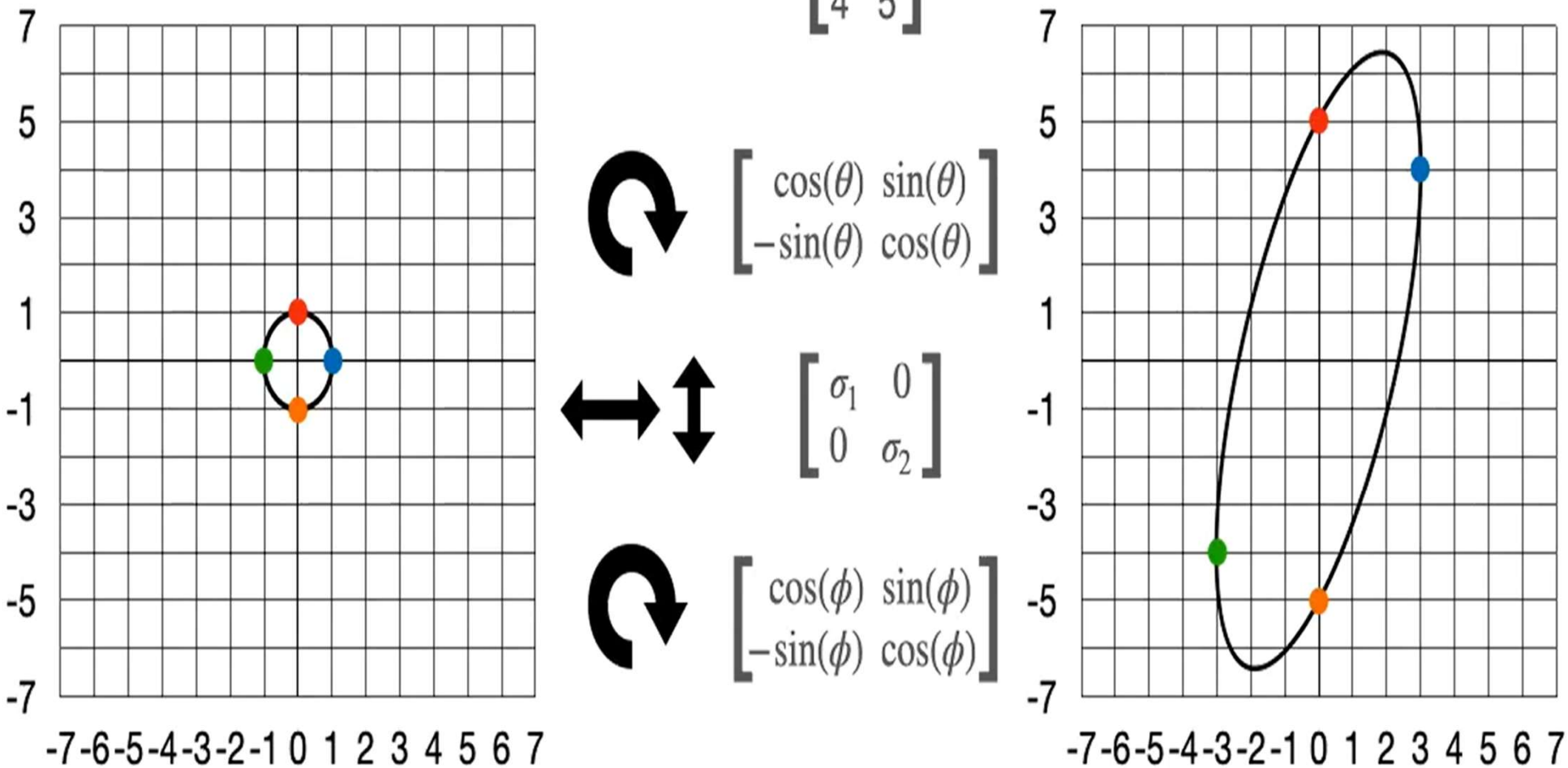
$$\begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$



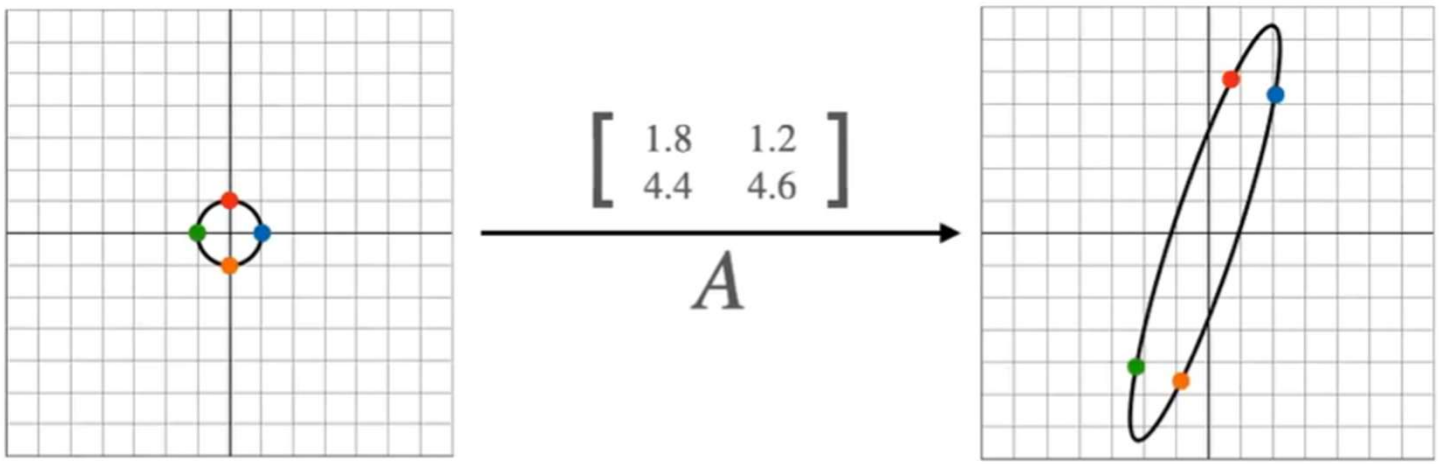


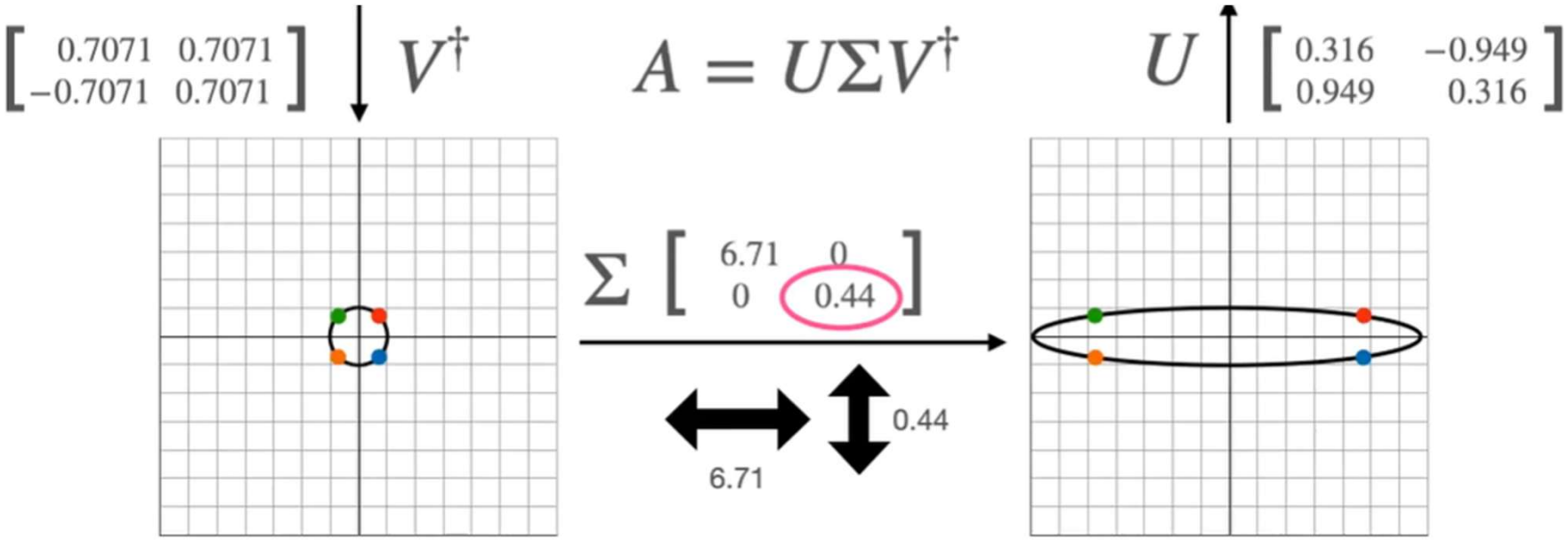
$$A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$$



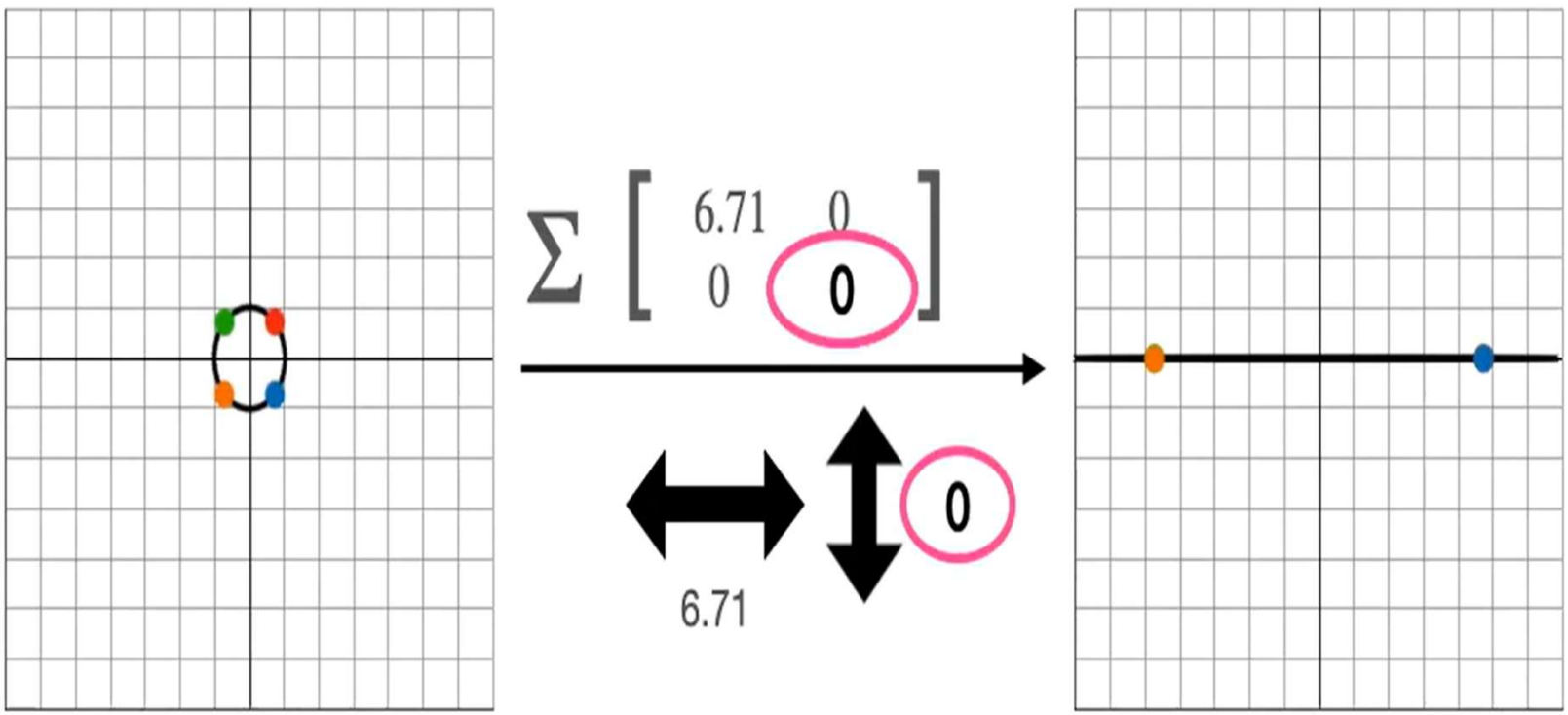
$$\begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{bmatrix}$$

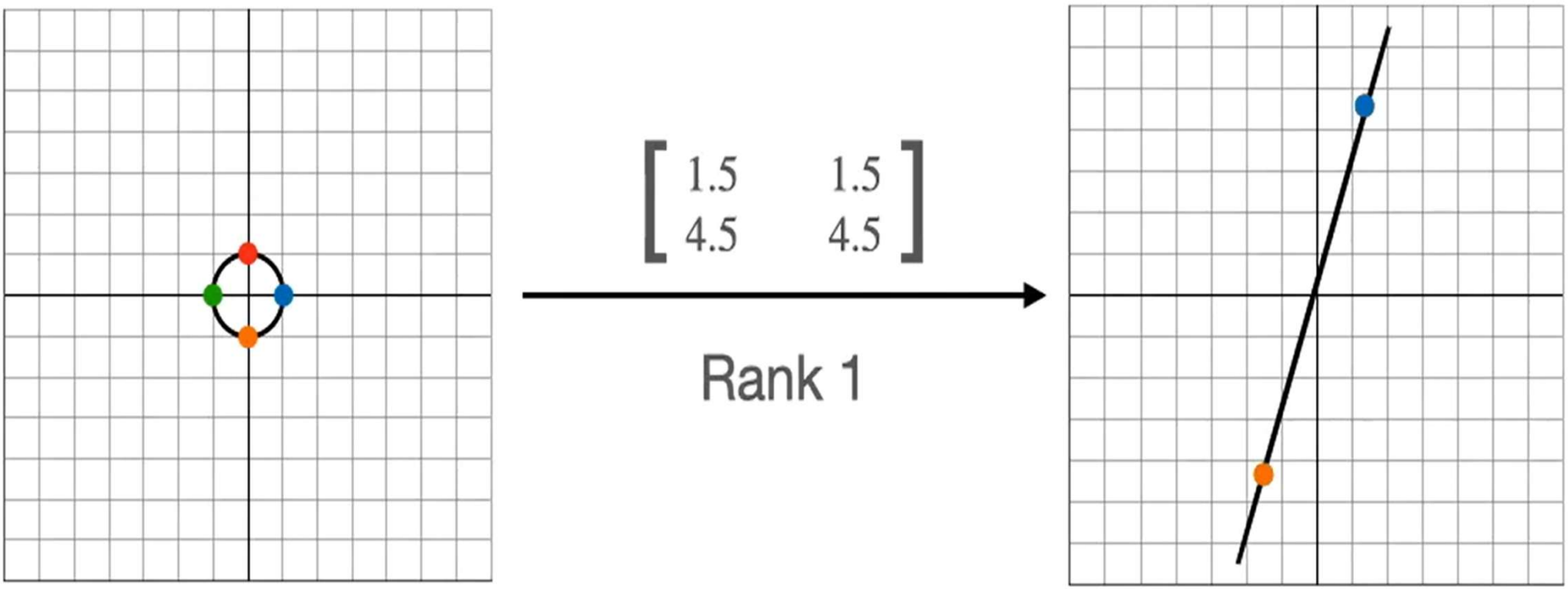
$$A = U \Sigma V^{\dagger}$$

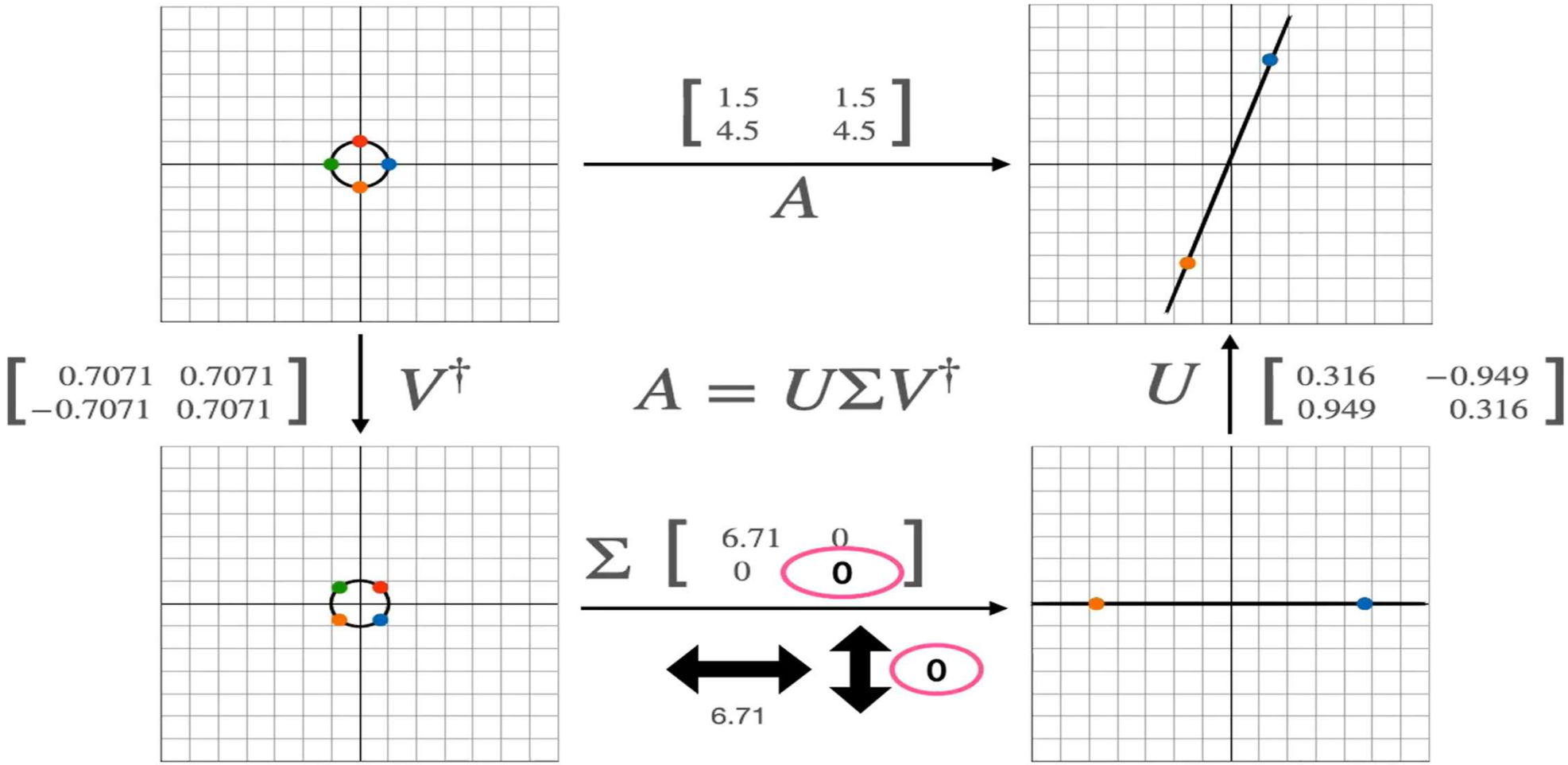

1.8 1.2
4.4 4.6
A

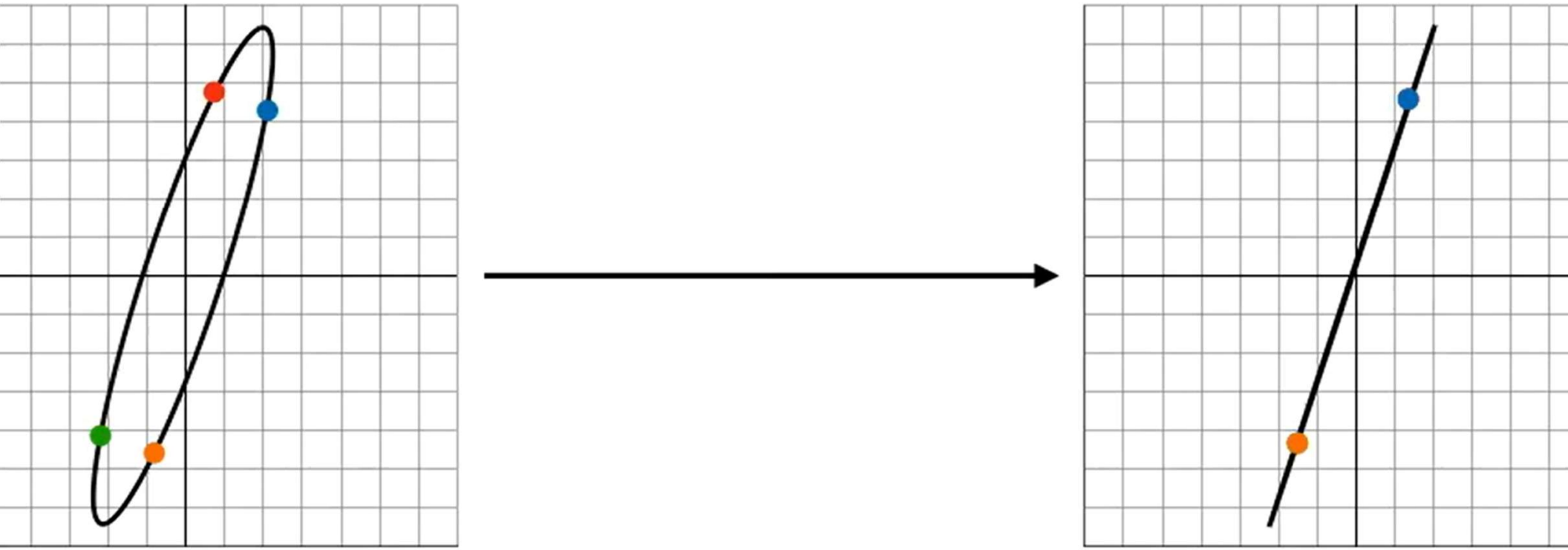

$\begin{bmatrix} 0.7071 & 0.7071 \\ -0.7071 & 0.7071 \end{bmatrix}$
$V^\dagger$
$A = U\Sigma V^\dagger$
U
$\begin{bmatrix} 0.316 & -0.949 \\ 0.949 & 0.316 \end{bmatrix}$
$\Sigma \begin{bmatrix} 6.71 & 0 \\ 0 & 0.44 \end{bmatrix}$
0.44
6.71

$$\begin{bmatrix} 0.7071 & 0.7071 \\ -0.7071 & 0.7071 \end{bmatrix} \downarrow V^{\dagger} \qquad A = U\Sigma V^{\dagger} \qquad U \left[\begin{bmatrix} 0.316 & -0.949 \\ 0.949 & 0.316 \end{bmatrix} \right.$$




1.5 1.5
4.5 4.5
Rank 1


$\begin{bmatrix} 1.5 & 1.5 \\ 4.5 & 4.5 \end{bmatrix}$
A
$\begin{bmatrix} 0.7071 & 0.7071 \\ -0.7071 & 0.7071 \end{bmatrix}$
$V^\dagger$
$A = U\Sigma V^\dagger$
U
$\begin{bmatrix} 0.316 & -0.949 \\ 0.949 & 0.316 \end{bmatrix}$
$\Sigma \begin{bmatrix} 6.71 & 0 \\ 0 & 0 \end{bmatrix}$
0
6.71
0



$$\begin{bmatrix} 1.8 & 1.2 \\ 4.4 & 4.6 \end{bmatrix}$$

Rank 2

$$\begin{bmatrix} 1.5 & 1.5 \\ 4.5 & 4.5 \end{bmatrix}$$

Rank 1

Rank 1 matrices

1	2	3	4
-1	-2	-3	-4
2	4	6	8
10	20	30	40

	1	2	3	4
1	1	2	3	4
-1	-1	-2	-3	-4
2	2	4	6	8
10	10	20	30	40

1	2	3	4
-1	-2	-3	-4
2	4	6	8
10	20	30	40

=

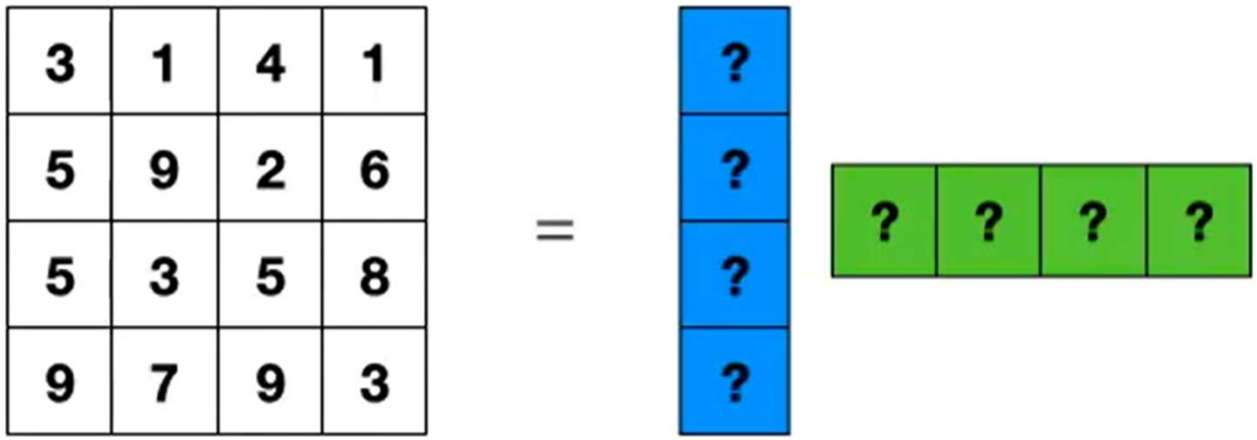
1
-1
2
10

1	2	3	4

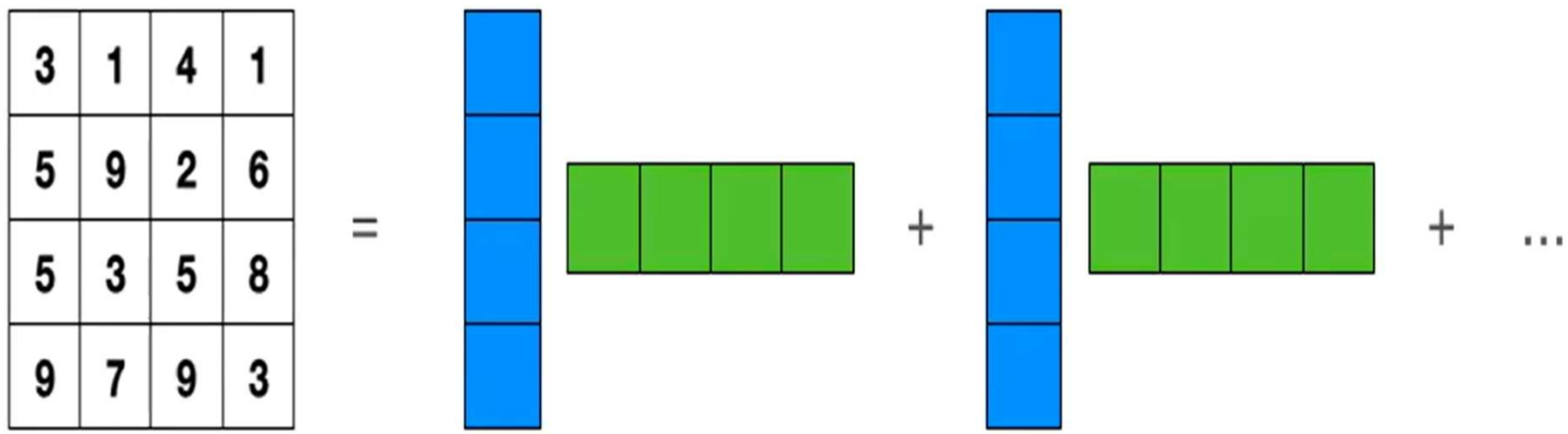
16 numbers

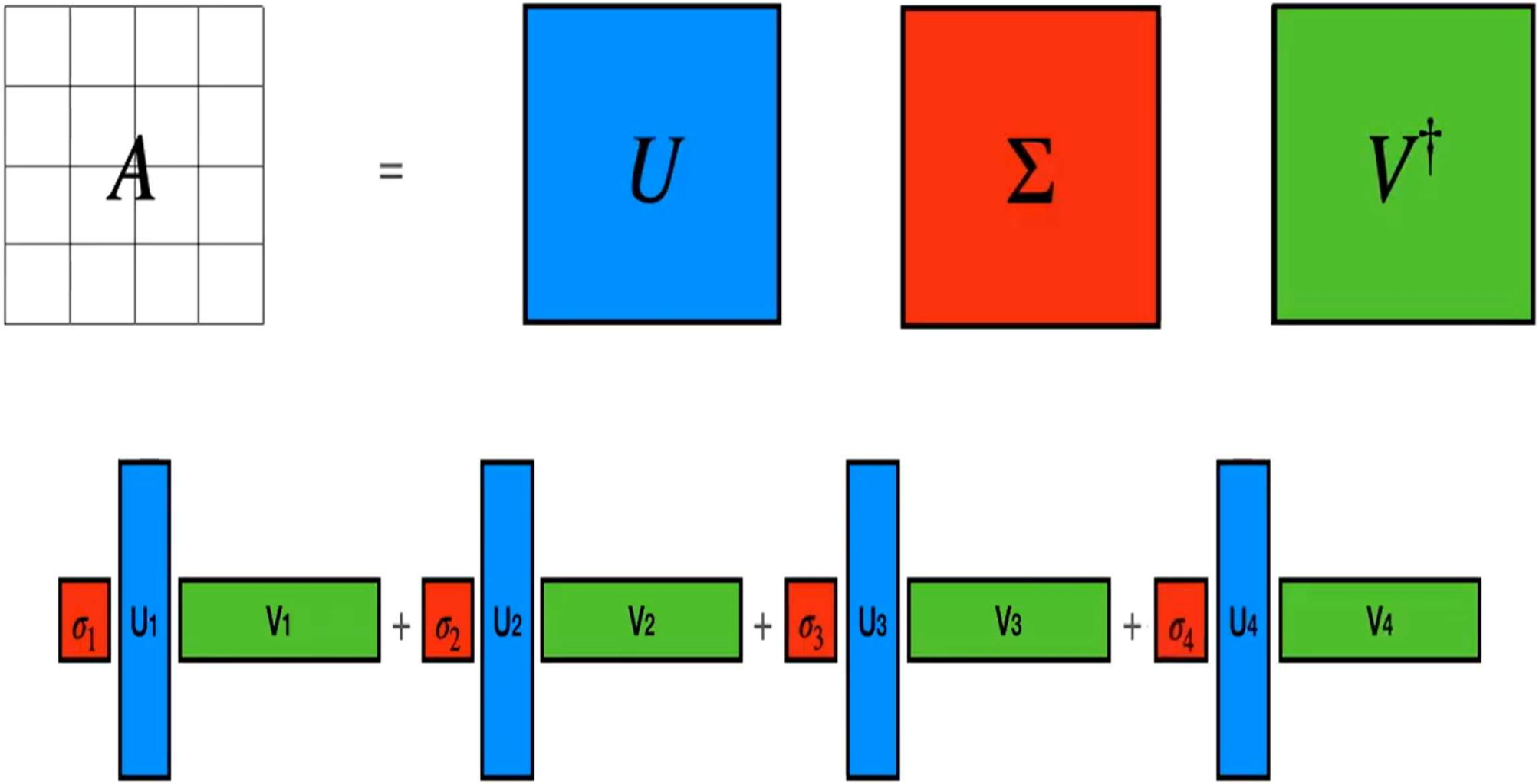
8 numbers

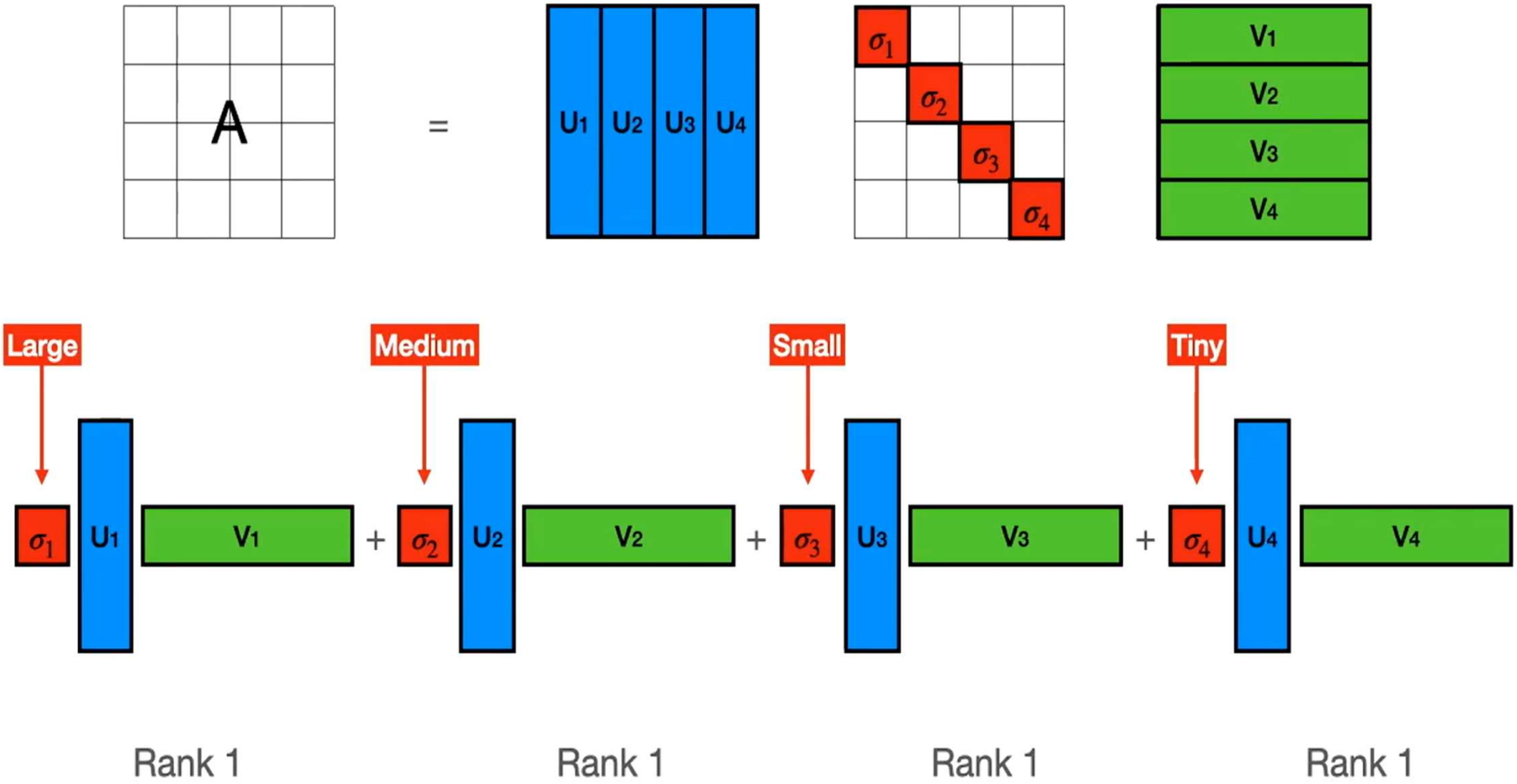
High rank matrix



Approximation by rank one matrices




A
=
U
Σ
V†
σ1
U1
V1
+
σ2
U2
V2
+
σ3
U3
V3
+
σ4
U4
V4


A
=
U1 U2 U3 U4
σ1 σ2 σ3 σ4
V1 V2 V3 V4
Large
Medium
Small
Tiny
σ1 U1 V1 + σ2 U2 V2 + σ3 U3 V3 + σ4 U4 V4
Rank 1
Rank 1
Rank 1
Rank 1

3	1	4	1
5	9	2	6
5	3	5	8
9	7	9	3

=

-0.21	0.37	-0.13	-0.89
-0.52	-0.7	0.43	-0.23
-0.48	-0.21	-0.84	0.15
-0.67	0.57	0.31	0.36

21.2			
	6.4		
		4.9	
			0.15

-0.55	-0.52	-0.49	-0.43
0.26	-0.4	0.65	-0.59
0.07	0.7	-0.22	-0.68
0.79	-0.29	-0.54	-0.04

Large

21.2	-0.21	-0.52	-0.48	-0.67	-0.55 -0.52 -0.49 -0.43

+

Medium

6.4	0.37	-0.7	-0.21	0.57	0.26 -0.4 0.65 -0.59

+

Small

4.9	-0.13	0.43	-0.84	0.31	0.07 0.7 -0.22 -0.68

+

Tiny

0.15	-0.89	-0.23	0.15	0.36	0.79 -0.29 -0.54 -0.04

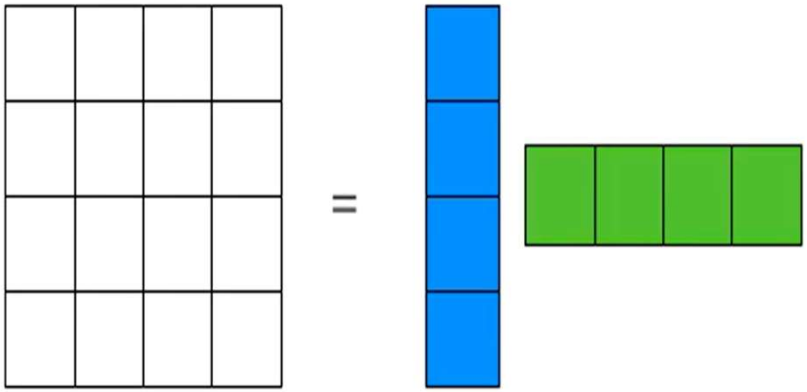
2.51	2.37	2.22	1.97
6.07	5,72	5.37	4.77
5.63	5,31	4.99	4.43
7.88	7.43	6.98	6.19

3.15	1.41	3.79	0.56
4.87	7.53	2.44	7.43
5.28	5.85	4.12	5.22
8.85	5.96	9.36	4.03

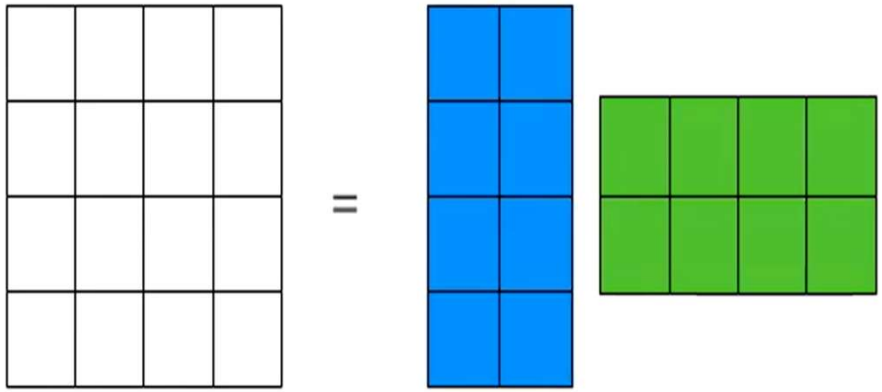
3.1	0.96	3.93	0.99
5.03	8.99	1.98	6
4.98	3.01	5.01	8
8.96	7.02	9.03	3

3	1	4	1
5	9	2	6
5	3	5	8
9	7	9	3

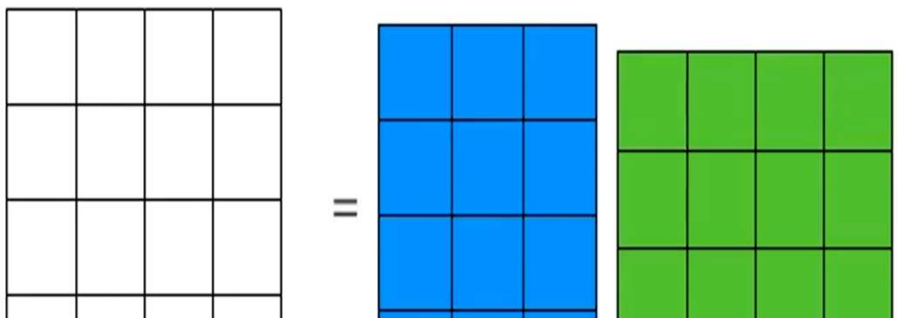
Rank of a matrix

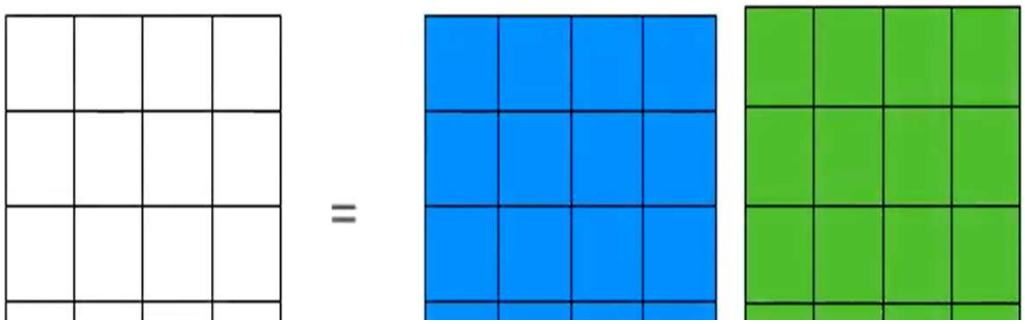


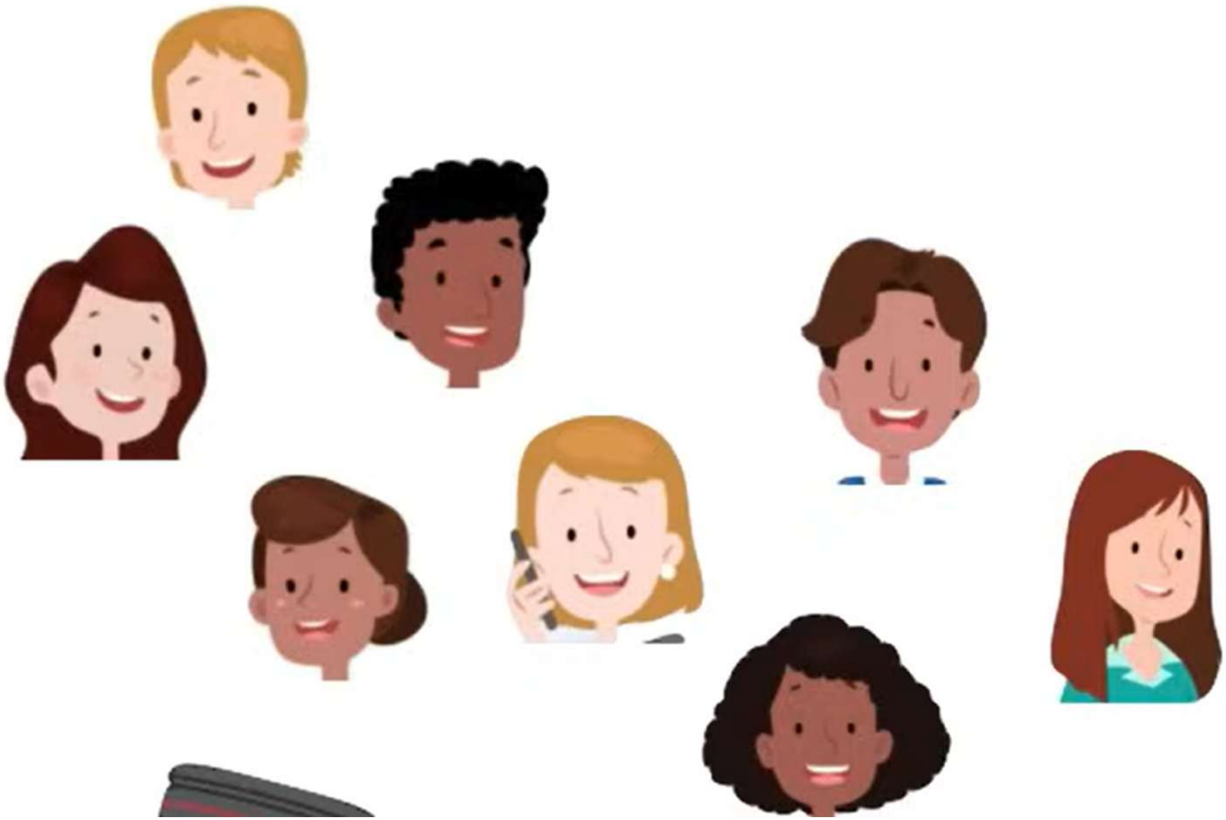
Rank 1

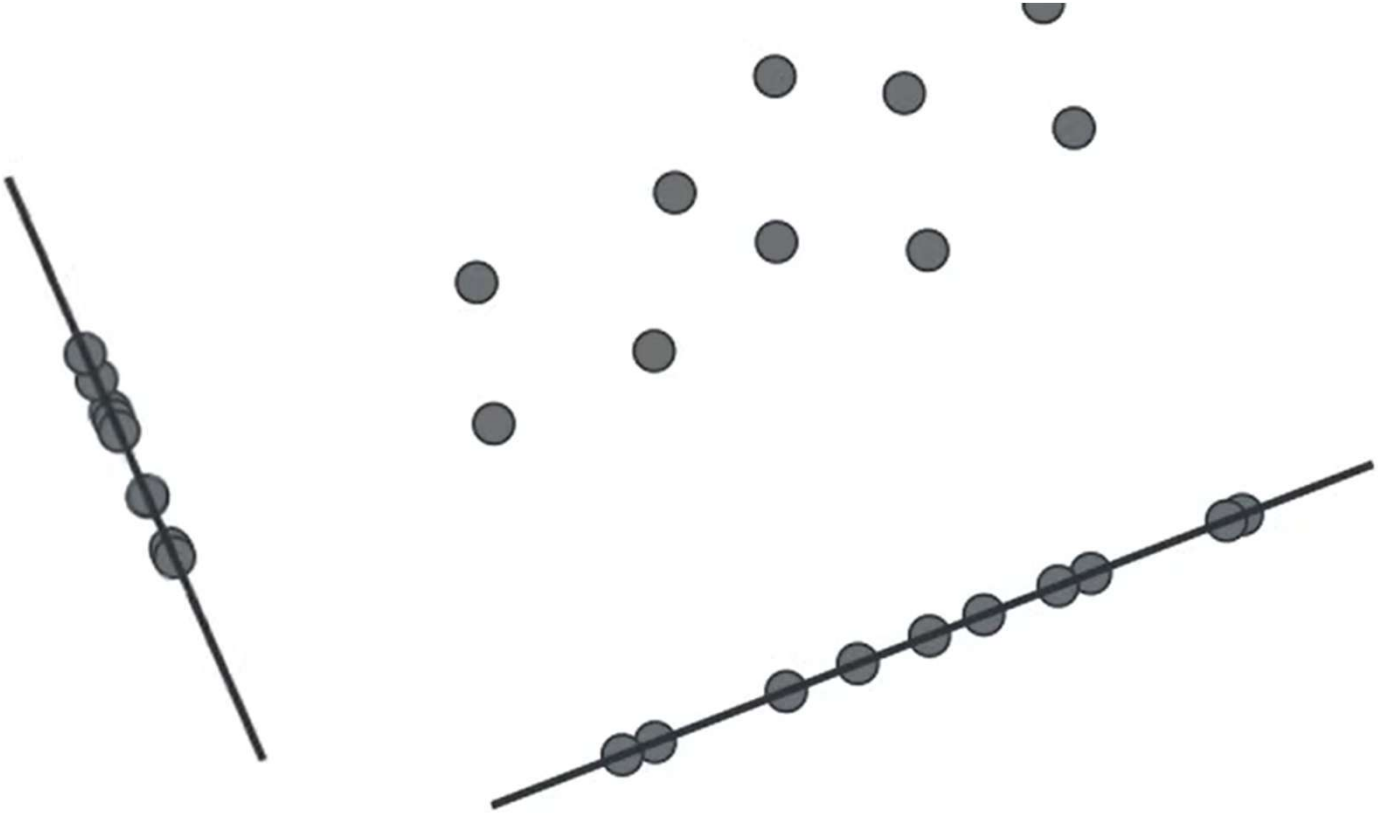
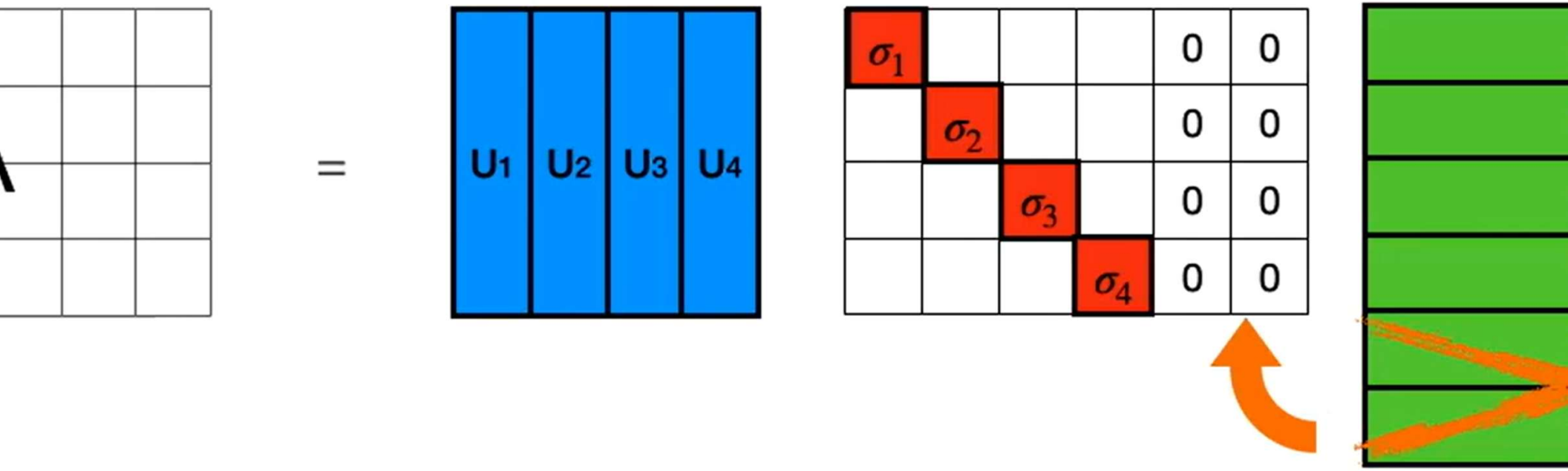


Rank 2









uare matrix? No problem!